A research submission for the ProveIt repository

Factorial fixed points

and smaller Diophantine consistency equations

Explicit quartics, complete coefficient files,
and shorter small-variable circuits

Prepared for Vladimir Reshetnikov
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Abstract

We give several completely materialized polynomial equations whose absence of solutions is equivalent in ZFC to Con(ZFC), by composing new arithmetic equivalences with the repository's identified proof-calculus calibration. The principal new lemma characterizes $n!$ as the unique fixed point

$$F = \left\lceil \frac{(F^2+1)^n}{\binom{F^2+n}{n}} \right\rceil, \qquad n, F \in \mathbb{N}.$$

It permits a proposed factorial value to supply its own precision bound. One square replaces a separate variable-exponent radius certificate. Together with a smaller local rule certificate and exact sign inference, this gives a quartic in 956 unrestricted integer variables with 510 quadratic residuals, 10 988 monomials, and explicitly listed coefficients. A companion in 989 integers has 7 705 monomials and coefficient height below 2^{26}. Natural-domain variants use 467, 468, or 469 variables. Independent improvements give a 456-operation nine-positive-variable equation of degree 1 376 378 and shorter exact radical norms. The archive contains all literal equations, proofs, generators, independent algebra checkers, source pins, and verification receipts. No global minimum, worldwide priority, proof of consistency, or new end-to-end proof-assistant certificate is asserted.

Comparison snapshot: `8f8fac484b976ba0ab75fd173282e838ca5a3b85`.

All variable counts refer to the domain stated next to the equation.

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1 What has been constructed

There are several reasonable meanings of "simplest": few unknowns, low degree, a short arithmetic circuit, few expanded terms, small coefficients, or a short semantic proof. The delivered equations improve several of these measures, but they do not order all the measures at once. The main construction follows the direct C222 proof calculus. It uses neither a simulation of the K215 Turing machine nor Jones' universal polynomial. The separate small-variable improvements retain their own parent source encodings and are identified as such.

Throughout, $\mathbb{N} = \{0, 1, 2, \ldots\}$, $\mathbb{Z}_{>0} = \{1, 2, \ldots\}$, and an unqualified integer variable ranges over all of $\mathbb{Z}$. The height of an expanded integer polynomial is the maximum absolute value of its collected coefficients. A quadratic residual system $r_1 = \cdots = r_s = 0$ specifies the *single* polynomial

$$P = \sum_{j=1}^{s} r_j^2. \tag{1}$$

The residuals are polynomial abbreviations, not additional unknowns. Every listed coordinate, including every retained arithmetic gate, is included in the variable count.

Theorem 1.1 (Main explicit integer equations)**.** *Let P_{956} be the polynomial specified by* `integer/equations/integer956/polynomial.json`*; equivalently it is the sum of the squares of the 510 literal residuals in the adjacent* `equation.json`*. It belongs to $\mathbb{Z}[X_0, \ldots, X_{955}]$, has exact degree four, 10,988 nonzero monomials, and height* $2\,445\,790\,974\,704$*. The ordinary mathematical proof in this submission, with its identified inherited calibration, proves in ZFC*

$$\mathrm{Con}(\mathrm{ZFC}) \quad \longleftrightarrow \quad \forall (x_0, \ldots, x_{955}) \in \mathbb{Z}^{956}, \quad P_{956}(x) \neq 0. \tag{2}$$

The companion P_{989} in `integer/equations/integer989/polynomial.json` *satisfies the same biconditional with 989 integer unknowns. It has 543 quadratic residuals, 7,705 monomials, exact degree four, and height* $63\,910\,295 < 2^{26}$*.*

The proof is constructive at the level of finite encodings: a zero recovers a finite ZFC refutation, and a refutation supplies a zero. Thus the more informative form of the result is

$$\exists p \in \mathbb{N}\ \ \mathrm{Proof}_{\mathrm{ZFC}}(p, \bot) \quad \longleftrightarrow \quad \exists x \in \mathbb{Z}^{956}\ \ P_{956}(x) = 0.$$

Negating this equivalence gives (2). No choice of coefficients depends on assuming consistency or inconsistency. No actual zero or proof of the absence of zeros is supplied.

1.1 The direct quartic tradeoffs

Table 1 compares the new direct family with the latest relevant incoming direct constructions at the pinned snapshot. The older 2,088-integer conversion is *not* used as the principal integer baseline: the incoming batched-certificate construction already has 1,690 integer variables. The parent and domain of each comparison matter.

The 467-variable natural equation uses the radius $H^2 F_H$; the 468- and 469-variable equations use F_H^2. The latter has a smaller raw parent and fewer expanded terms, although the deterministic elimination order leaves one more variable. This is a real tradeoff, so both complete variants are retained.

Relative to the 1,690-variable direct integer baseline, P_{956} removes 734 variables and 10,393 monomials. These are reductions of about 43.4% and 48.6%, respectively. It is not a claim that 956 is the smallest possible integer-variable count: the repository already contains much smaller variable counts with different degrees and coefficient representations.

Construction	Domain	Variables	Rows	Monomials	Height
Synchronized incoming	$\mathbb{N}$	522	400	20,225	83,900,628
New, output-linear radius	$\mathbb{N}$	467	363	18,479	83,880,005
New, square radius	$\mathbb{N}$	468	364	14,842	83,850,989
New, smaller support	$\mathbb{N}$	469	365	12,051	83,850,757
Batched incoming	$\mathbb{Z}$	1,690	842	21,381	2,649,643,805,758
New, fewer variables	$\mathbb{Z}$	956	510	10,988	2,445,790,974,704
New, smaller coefficients	$\mathbb{Z}$	989	543	7,705	63,910,295

Table 1: Direct literal equations. Every degree is exactly four; "rows" counts the quadratic residuals before the final squared sum.

1.2 Separate improvements at small variable counts

Domain	Variables	Degree	Parent operations	New operations
$\mathbb{Z}_{>0}$, original31	9	1,376,378	475	456
$\mathbb{Z}_{>0}$, compact54	9	1,520,186	433[a]	442
$\mathbb{Z}_{>0}$	11	211	111	110
$\mathbb{Z}$	13	1,408	185	176
$\mathbb{Z}$	13	1,392	186	177
$\mathbb{Z}$	13	1,088	195	186
$\mathbb{Z}$	13	1,072	195	186

Table 2: Live binary arithmetic operations. Addition, subtraction, multiplication, squares, and scalar multiplication are charged; register reuse is free. [a]The 433-operation parent has degree 1,628,768, so this row is a cost–degree tradeoff.

For the first row, the previous degree is 1 379 730: both cost and degree improve. For the last five rows the entire polynomial is unchanged; only its evaluator is shorter. In particular, the degree-1,072 result improves the latest incoming 195-operation degree-1,072 equation, rather than claiming its parent's degree reduction anew.

These arithmetic operation counts treat a fixed numeral as a leaf. The fixed leaves in the small-variable families can be very large. Their complete bytes and reconstruction bindings are included, but the cost measure is not a bit-operation or printed-digit complexity measure. The direct quartics instead provide ordinary, fully collected, modest-sized integer coefficients.

2 The factorial fixed point

The usual ceiling representation of a factorial chooses an independent radius large enough to make a rational approximation lie within one of $n!$. The new observation is that the unknown value can help select the radius, provided the resulting fixed point is proved unique. This removes an arithmetic graph rather than merely abbreviating its notation.

For $n, R \in \mathbb{N}$, define

$$f_n(R) = \frac{(R+1)^n}{\binom{R+n}{n}} = n! \prod_{j=1}^{n} \frac{R+1}{R+j}.$$

Empty products are one. The denominator is positive.

Lemma 2.1 (Approximation and monotonicity)**.** *For $n \geq 2$ and real $R > 0$, $f_n(R)$ is positive, nondecreasing in R, at most $n!$, and satisfies*

$$0 \leq n! - f_n(R) < \frac{n!\, n(n-1)}{2R}. \tag{3}$$

The formula for real R here means the finite product, so no generalized binomial coefficient is needed.

Proof. Each factor is positive and at most one. For $j \geq 1$, the function $(R+1)/(R+j)$ is nondecreasing, by cross multiplication (or differentiation). Put $u_j = (j-1)/(R+j)$. Then $0 \leq u_j < 1$, and induction on the finite product gives

$$1 - \prod_{j=1}^{n}(1-u_j) \leq \sum_{j=1}^{n} u_j < \frac{1}{R}\sum_{j=1}^{n}(j-1) = \frac{n(n-1)}{2R}.$$

The strict inequality holds because $n \geq 2$ gives a positive summand with denominator strictly greater than R. Multiply by $n!$. □

Theorem 2.2 (Square-radius fixed point)**.** *For all $n, F \in \mathbb{N}$,*

$$F = \left\lceil \frac{(F^2+1)^n}{\binom{F^2+n}{n}} \right\rceil \quad \longleftrightarrow \quad F = n!. \tag{4}$$

Proof. Let $K = n!$ and suppose first that $F = \lceil f_n(F^2) \rceil$. The quotient is positive and at most K, so $1 \leq F \leq K$. For $n \geq 4$, monotonicity and $F^2 \geq 1$ give

$$f_n(F^2) \geq f_n(1) = \frac{2^n}{n+1} > n - 1.$$

The last inequality is $2^n > n^2 - 1$. It holds at $n = 4$; its induction step follows from $2(n^2-1) - ((n+1)^2 - 1) = n^2 - 2n - 2 > 0$ for $n \geq 4$. Consequently $F \geq n$.

Apply (3) with $R = F^2$. First,

$$K - f_n(F^2) < \frac{Kn(n-1)}{2F^2} < \frac{K}{2},$$

so $F \geq f_n(F^2) > K/2$. Applying the same estimate a second time,

$$K - f_n(F^2) < \frac{Kn(n-1)}{2F^2} < \frac{2n(n-1)}{K} = \frac{2}{(n-2)!} \leq 1.$$

Thus $K - 1 < f_n(F^2) \leq K$, and its ceiling is K.

At $n = 0, 1$ the quotient is identically one. At $n = 2$ and $F \geq 1$,

$$f_2(F^2) = \frac{2(F^2+1)}{F^2+2} \in (1, 2),$$

so $F = 2$. At $n = 3$, the already established bound $1 \leq F \leq 6$ leaves six exact rational values:

F	1	2	3	4	5	6
$\lceil f_3(F^2) \rceil$	2	4	5	6	6	6.

Only $F = 6$ is a fixed point. This proves soundness in every case.

Conversely, set $F = K$. For $n \geq 2$, (3) gives

$$0 \leq K - f_n(K^2) < \frac{n(n-1)}{2K} = \frac{1}{2(n-2)!} \leq \frac{1}{2}.$$

Hence its ceiling is K. The cases $n = 0, 1$ are immediate. □

This is an unrestricted elementary proof, not an experimental fixed-point search. The bounded calculations in the audit scripts check transcription and small cases; they are not the reason the theorem holds for arbitrary n.

2.1 A common radius for several factorials

Corollary 2.3 (Shared radius)**.** *Suppose $H \geq 1$, the first ceiling certificate uses the argument H and radius $R = F_H^2$, and every later argument n satisfies $0 \leq n \leq H$. Then the first certificate forces $F_H = H!$, and the same radius correctly represents every later factorial by $F_n = \lceil f_n(R) \rceil$.*

Proof. Theorem 2.2 handles the first certificate. For $2 \leq n \leq H$, with $R = (H!)^2$,

$$0 \leq n! - f_n(R) < \frac{n!\, n(n-1)}{2(H!)^2} \leq \frac{H(H-1)}{2H!} = \frac{1}{2(H-2)!} \leq \frac{1}{2}.$$

The cases $n = 0, 1$ have quotient one, and if $H = 1$ these are the only cases. Completeness uses the actual factorial values and their remainders. □

The concrete compiler requests $H!$ first and $t!$ second, with $H \geq t \geq 1$. It uses one square F_H^2 for both. That same square already occurs in the base-modulus equation for the power request F_H^t, so it is shared in the final arithmetic circuit.

2.2 The exact polynomial ceiling graph

Let N, C be positive naturals and let F, r, s be naturals. Then

$$N + C = CF + r + 1, \qquad C = r + s + 1 \tag{5}$$

is equivalent, existentially in r, s, to $F = \lceil N/C \rceil$. Indeed the second row gives $0 \leq r < C$, and the first gives $(F-1)C < N \leq FC$. Conversely these strict and weak inequalities provide $r = N + C - CF - 1 \in [0, C-1]$ and $s = C - r - 1$.

In the factorial graph, $C = \binom{R+n}{n}$ and $N = (R+1)^n$. Their notation describes the semantics of separate finite polynomial graphs. No ceiling, factorial, binomial symbol, or variable exponent remains in `equation.json` or `polynomial.json`.

2.3 The retained output-linear alternative

An especially short fixed-point proof uses $R = n^2 F$:

$$F = \left\lceil f_n(n^2 F) \right\rceil \quad \longleftrightarrow \quad F = n!. \tag{6}$$

For $n \geq 2$, a proposed fixed point has $1 \leq F \leq K = n!$, and

$$K - f_n(n^2 F) < \frac{K(n-1)}{2nF} < \frac{K}{2F}.$$

Initially $F \geq 1$ forces $f_n(n^2F) > K/2$, hence $F > K/2$. The same inequality then gives defect less than one, proving soundness. For $F = K$ the defect is less than $1/2$, proving completeness. The cases $n = 0, 1$ are immediate. With a dominant argument H, the shared radius $H^2F_H = H^2H!$ gives, for $2 \leq n \leq H$, a defect strictly below $(H-1)/(2H) < 1/2$.

This alternative produces a 783-variable parent and the 467-variable natural contraction. The square-radius version has a 781-variable parent and better term counts. Both are compile-time choices of a single literal polynomial, not choices made by solutions of the equation.

3 A smaller direct certificate for the proof rules

The source is the inherited numerical calculus C222, with truth label 0 and target falsity label 1. Its numerical constructors include

$$\begin{aligned} I(a,b) &= 2(a+b)(a+b+1)+4b, & A(a,b) &= I(a,b)+2,\\ M(a,b) &= I(a,b)+1, & E(a,b) &= I(a,b)+3,\\ N(a) &= I(a,1), & X(a,b) &= N(A(a,N(b))). \end{aligned}$$

These are literal polynomials. Their complete axiom instances occur in `direct/code/hybrid_source.py`, and the constructor prefix is checked by exact syntax comparison with the inherited source. The calculus contains truth, eight unconditional axiom schemata, one guarded bundle schema, modus ponens, and generalization.

3.1 Eliminate premise labels inside each rule branch

Earlier local certificates carried independent premise labels p, q even though each selected branch determines them. The new bounded row uses just nine coordinates

$$(v, a, b, c, d, S, S', u, u').$$

Here v is the conclusion, a, b, c, d are schema parameters, S, S' are successive accumulator values, and u, u' are beta quotients. Let $h_1, \ldots, h_8$ be the eight inherited unconditional axiom polynomials and h_B the bundle polynomial. Put

$$\begin{aligned} G &= ((c-a)(c-b))^2 - (d+1)^2,\\ K &= v\prod_{j=1}^{8}(v-h_j), & B &= (v-h_B)^2+G^2,\\ w(z) &= 1+z\mathcal{M}, & \mathcal{M} &= M_0+1. \end{aligned}$$

Define the rule residuals and nonnegative branch tests by

$$\begin{aligned} L_A &= S'w(v) - S, & Q_A &= K^2B + L_A^2,\\ L_M &= S'w(v) - Sw(a)w(I(a,v)), & & \\ L_G &= S'w(v) - Sw(b), & Q_G &= (v-A(a,b))^2 + L_G^2. \end{aligned}$$

The local polynomial at index i is

$$\begin{aligned} P_{\mathrm{loc}} = {} & Q_A L_M^2 Q_G\\ & + [x - S - (1+(i+1)y)u]^2\\ & + [x - S' - (1+(i+2)y)u']^2. \end{aligned} \tag{7}$$

Every displayed branch factor is nonnegative, including on real inputs.

If $Q_A = 0$, the conclusion is truth or one of the valid axiom instances; restore $p = q = 0$. If $L_M = 0$, restore $p = a$, $q = I(a, v)$: this is modus ponens with the correct accumulator equation. If $Q_G = 0$, restore $p = b$, $q = 0$, with $v = A(a, b)$: this is generalization. Conversely each valid old branch has this parameterization. Unused parameters can be zero. The proof makes no claim that parameters satisfy unselected branches. Existential quantification distributes over the finite disjunction represented by the product.

The guard $G = 0$ says $|(c - a)(c - b)| = d + 1 > 0$, exactly the required pair of disequalities. Its natural witness exists whenever those disequalities hold. The branch-specific substitution therefore preserves the actual Horn rule relation, not merely some numerical evaluations of its source.

3.2 Restore the correct carrier bound

Bound v, a, b, c, d by $m \geq 1$. The restored second premise in modus ponens need not be bounded by m; it is bounded by $I(m, m)$. Hence use the label bound

$$L(m) = I(m, m) + 2 = 8m^2 + 8m + 2 = 2(2m + 1)^2$$

and impose

$$\mathcal{M} = (L(m) + 1)^t, \qquad t \geq 1. \tag{8}$$

The inherited bounded-factor carrier theorem applies with $L(m)$. It converts the product identity at this sufficiently large carrier into the formal label-multiset identity used by the finite Horn-flow theorem. All active labels are in its stated bound; large evaluations in inactive axiom branches need not be label-bounded.

The remaining local bounds are $S, S' \leq y$ and $u, u' \leq x$. They make the two beta reconstruction rows in (7) have strict remainder bounds automatically. The endpoints are

$$x = M_0 + 2 + (y + 1)u_0, \qquad y = M_0 + r_0 + 2,$$
$$x = 1 + [1 + (t + 1)y]u_t.$$

Thus the initial accumulator is $\mathcal{M} + 1$ (target label one), the final accumulator is one, and the beta code links consecutive accumulators. Their product telescopes to the certified Horn flow.

Given a finite proof, first choose sufficiently large t, m, then the carrier, the finite accumulator path, and beta data y, x. These are ordinary finite objects. No uniform bound on a possible ZFC refutation is assumed or needed.

3.3 The explicit majorant

Split the signed arithmetic circuit for P_{loc} into nonnegative-coefficient circuits $E - F$. Addition, subtraction, and multiplication on pairs use the exact ring identities

$$(e, f) + (g, h) = (e + g, f + h), \quad (e, f) - (g, h) = (e + h, f + g),$$

$$(e, f)(g, h) = (eg + fh, eh + fg).$$

Evaluate at the index bound t and coordinate bounds $(m, m, m, m, m, y, y, x, x)$. The actual stored DAG has a checked power majorant exponent 6 893 174. Set

$$H = (2 + t + x + y + m + M_0)^{2^{23}}. \tag{9}$$

Since $2^{23} = 8\,388\,608$ is larger, H bounds both sides, the index, and all local bounds with the required strict margin. Equation (9) takes exactly 23 successive squarings.

The certificate checks this bound gate by gate. If the base is $R \geq 2$, input atoms are at most R, a sum of quantities bounded by R^a, R^b is at most $R^{\max(a,b)+1}$, and a product is at most R^{a+b}. Constants are bounded by a suitable fixed power of two. No semantic claim about a refutation is hidden in this numerical majorant.

4 Inherited compiler contracts and the consistency endpoint

The new projection and factorial argument are composed with the following specific arithmetic interfaces. All variables in this section range over $\mathbb{N} = \{0, 1, \ldots\}$ unless indicated otherwise. For the unrestricted integer equation, the sign-recovery certificate must first establish this natural domain. The statements below concern every satisfying assignment; their use does not presume the intended values of intermediate witnesses.

4.1 Logical calibration and what is imported

The numerical source is the fixed twelve-branch calculus C_{222}, with truth code 0 and falsity code 1. Its calibration is

$$C_{222}(1) \iff \mathrm{BProv}(\mathrm{ZF}, \bot) \iff \mathrm{BProv}(\mathrm{ZFC}, \bot).$$

The exact named results are `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. Their frozen files, relative to the direct-equation component, are `provenance/upstream/calibration/KEndpoint.lean` and `provenance/upstream/calibration/ConstructibleChoice.lean`. The source numerals and rules are in the adjacent `Code254.lean` and `KCode222.lean`. The repository paths are `Computability/BusyBeaver/Lean/BusyBeaver/ZFC/`, at calibration commit `08200eb10fde2db408c69dffd840d6920a569b1f`; exact file hashes are retained in the provenance ledgers.

These are substantial imported mathematical results. The first combines the soundness of the numerical calculus with its model-existence completeness argument; the second removes Choice by the constructible-model argument and first-order soundness/completeness. No Turing-machine endpoint is used. The appeal is to these theorems and their proofs formalizable in ZFC, not to an inference that every external Lean theorem is automatically a theorem of ZFC. The complete transitive Lean development was not rebuilt for this contribution, and no new object-language ZFC proof term is claimed.

The remaining passage from finite rule instances to a proof is the single-conclusion Horn-flow criterion. Give $(p, q) \to v$ the vector $e_v - e_p - e_q$, with $e_0 = 0$. A proof tree has total flow e_1. Conversely, if a finite nonnegative sum of rule vectors equals e_1, sum coordinates outside the least rule-closed set containing 0. Each rule contributes at most zero outside that set; a missing target would contribute one, a contradiction. Cycles do not invalidate this argument.

For the accumulator-to-flow step, all restored labels lie in $[0, L]$, where $L = 8m^2 + 8m + 2$, and $M = (L+1)^t$, $t \geq 1$. Telescoping the accumulator equations with endpoints $S_0 = 1 + M$, $S_t = 1$ gives $P(M) = Q(M)$, where

$$P(X) = \prod_{i<t}(1 + v_i X), \qquad Q(X) = (1 + X)\prod_{i<t}(1 + p_i X)(1 + q_i X).$$

Since $P(M) < M^{t+1}$, nonnegativity first forces $\deg Q \le t$. Both products then have at most t nonconstant factors, so every coefficient is strictly below M. The least nonzero coefficient of $P - Q$ could not vanish on evaluation at M; hence $P = Q$. Unique factorization in $\mathbb{Q}[X]$ gives precisely the required flow. This is the bounded-factor argument in `provenance/upstream/calibration/direct_flow_large_carrier_README.md`. Conversely, expand a proof tree and let S_i be the product of the weights of its open obligations. A finite CRT beta code gives $x = S_i + (1 + (i+1)y)u_i$, choosing y larger than every state and divisible by $1, \dots, t+1$. In soundness, the supplied bounds $S_i \le y < 1 + (i+1)y$ make these actual remainders and make adjacent rows agree. Thus this encoding is an equivalence of existence predicates.

4.2 The bounded-row CRT contract

Let E, F be polynomials with nonnegative integer coefficients in the row index, fixed outer parameters, and r local coordinates. In the present source $r = 9$ and the coordinate bounds are

$$\mathbf{u} = (m, m, m, m, m, y, y, x, x).$$

Suppose $t \ge 1$ and

$$H \ge E(t, \mathbf{u}) + F(t, \mathbf{u}) + t + x + y + m + 1.$$

With the outer parameters suppressed, put $T = H!$, $d_i = 1 + (i+1)T$, and $\mathcal{B} = \prod_{i<t} d_i$. Then the bounded family

$$\forall i < t \; \exists w \le \mathbf{u} : \; E(i, w) = F(i, w)$$

is equivalent to the existence of natural $C, A_1, \dots, A_r$ such that

$$\begin{aligned} &\mathcal{B} \mid \binom{A_j}{u_j + 1} && (1 \le j \le r), \\ &\mathcal{B} \mid 1 + (C+1)T, && E(C, A) \equiv F(C, A) \pmod{\mathcal{B}}. \end{aligned}$$

The representatives A_j may all be required positive, as they are in the literal compiler.

For soundness, choose a prime $\ell \mid d_i$. Since d_i is coprime to $H!$, necessarily $\ell > H$. Multiplying the range congruence by $(u_j + 1)!$ gives a zero falling factorial modulo ℓ, so $A_j \equiv w_j \pmod{\ell}$ for some $0 \le w_j \le u_j$. The index congruence gives $C \equiv i \pmod{\ell}$ by cancellation of T. The two evaluations are congruent and both lie in $[0, H]$, hence are equal. For completeness, the d_i are pairwise coprime: a common prime would divide $j - i$ and so divide $H!$, which is impossible. CRT encodes the chosen row witnesses. Each $(u_j + 1)!$ divides $H!$ and is coprime to $\mathcal{B}$, so it cancels from the falling-factorial congruence. Positive representatives follow by adding multiples of $\mathcal{B}$. The literal evaluation congruence uses $E + \mathcal{B}q = F + \mathcal{B}v$ with natural q, v; arbitrary signed quotients are represented by choosing the common offset v sufficiently large, so this orientation adds no restriction.

The supplied majorant certificate checks 392 gate inequalities and bounds $E+F+t+x+y+m+1$ by $R_0^{6,893,174}$, where $R_0 = 2 + t + x + y + m + M_0 \ge 2$. The literal choice $H = R_0^{2^{23}}$ is therefore sufficient. Each addition uses the exponent bound $\max(e_1, e_2) + 1$, each multiplication uses $e_1 + e_2$; induction on the actual nonnegative pair circuit proves the bound. The factorial bootstrap supplies the genuine value of T before this CRT theorem is invoked.

The fixed-length modulus product is itself represented arithmetically. For $Q = (1 + (t+1)T)^t + 2$ and $D = TQ - 1 > 0$,

$$\mathcal{B} = \left(T^t \, t! \binom{Q + t}{t} \right) \bmod D.$$

Indeed the numerator is $\prod_{i=1}^{t}(TQ+iT)$, congruent to $\prod_{i=1}^{t}(1+iT)$ modulo D. The latter product is strictly smaller than D by the defining bound on Q. An ordinary quotient/remainder equation with remainder slack supplies this graph; no unbounded product operator remains in the polynomial.

4.3 One radix for all binomial requests

For the twelve actual requests $\binom{n_j}{k_j}$, all n_j, k_j are positive. Let N be the sum of the distinct upper-argument circuit atoms and $q = 2^N + 1$. Thus $q > 2^{n_j}$. With $L_j = q^{k_j}$ and $U_j = (q+1)^{n_j}$, the literal natural equations are

$$U_j = L_j(qh_j + C_j) + r_j, \qquad L_j = r_j + s_j + 1, \qquad q = C_j + d_j + 1.$$

They recover exactly digit k_j of the carry-free base-q expansion of $(q+1)^{n_j}$; therefore $C_j = \binom{n_j}{k_j}$, including $k_j > n_j$. Conversely ordinary division supplies all four natural auxiliary values. This theorem is simultaneous for arbitrary assigned argument values. In completeness, the shared radix is selected after all semantic upper arguments, including the chosen CRT representatives, have been fixed.

4.4 Shared Pell certificates for all positive powers

Write $(a+\sqrt{a^2-1})^j = X_j(a) + Y_j(a)\sqrt{a^2-1}$ for $a \geq 2$. The imported number-theoretic ingredients are classification of natural Pell solutions and the finite Matiyasevich index criterion. The exact repository interface is `DiophantineConsistency.PellRepresentation.criterion_iff_pell`, in `provenance/upstream/calibration/PellRepresentation.lean`, which invokes `Pell.matiyasevic` from Mathlib's `Mathlib/NumberTheory/PellMatiyasevic.lean`.

For the 23 distinct power requests (n, k, m), the common frame is

$$W = 2 + \sum\{\text{distinct base, exponent, and output atoms}\}, \quad Z = z + 1, \quad A^2 - W^3(W+2)Z^2 = 1.$$

Each of the seven distinct bases has a natural T_n satisfying $2An = T_n + n^2 + 1$. This equation excludes $n = 0$ before any factorial or power value is recovered. Every exponent is syntactically positive. Pell classification applied at base $W+1$ gives $A = X_j(W+1)$ and $WZ = Y_j(W+1)$. Modulo W, the recurrences give $Y_j(W+1) \equiv j$; thus $j \geq W$. Since $1 \leq n, k \leq W < A$ and $m < W$, consequently

$$A \geq (W+1)^W > W^W, \qquad T_n \geq 2A - 2 > W^W \geq n^k, \qquad m < T_n.$$

Conversely $j = W$ supplies a frame for any finite family of intended values. The weaker inequality $A > W^2$ alone is insufficient for power recovery; the divisibility of the frame's Pell index is essential.

For each of the eighteen exponent atoms k_i, use natural $X_i, Y_i, S_i, R_i, e_i, s_i, r_i$. The literal shared index equations are

$$\begin{aligned}
Y_i &= k_i + e_i, & X_i^2 - (A^2-1)Y_i^2 &= 1,\\
S_i^2 - (B^2-1)R_i^2 &= 1, & S_i &= X_i + Us_i,\\
R_i &= k_i + 4Y_i r_i, & U^2 - (A^2-1)V^2 &= 1,\\
V &= \left(\prod_i Y_i^2\right)(v_0+1), & B &= 1 + 4V(b_0+1) = A + Ub_1.
\end{aligned}$$

They imply every hypothesis of the nonzero branch of the imported index criterion: $A, B > 1$, $Y_i \geq k_i > 0$, $V > 0$, $Y_i^2 \mid V$, and the four required congruences. Hence $(X_i, Y_i) = (X_{k_i}(A), Y_{k_i}(A))$. For simultaneous completeness, set $L = 4\prod_i Y_i^2$ and choose a positive power of the unit $A + s$ equal to one in the finite ring $(\mathbb{Z}/L\mathbb{Z})[s]/(s^2 - (A^2 - 1))$. Its Pell coordinates give $U \equiv 1 \pmod{L}$, $V \equiv 0 \pmod{L}$, $V > 0$, and $\gcd(U, 4V) = 1$. CRT chooses $B > A$ with the displayed congruences. Set $S_i = X_{k_i}(B)$, $R_i = Y_{k_i}(B)$; recurrence congruences and monotonicity give all required natural quotients.

Finally each request has one natural quotient h and

$$X_i + nY_i = AY_i + m + T_n h.$$

For $H_j = X_j(A) - (A - n)Y_j(A)$, the recurrence is $H_{j+1} = nH_j + T_n Y_j(A)$, $H_0 = 1$. Thus

$$H_k - n^k = T_n \sum_{j<k} n^{k-1-j} Y_j(A).$$

Both m and n^k are in $[0, T_n)$, proving soundness; the displayed nonnegative sum supplies h for completeness. The finite-ring and recurrence arguments justify shared witnesses; merely identifying independently existential variables would not suffice.

Composition inside ZFC. These proofs use finite induction, integer/rational arithmetic, prime factorization, CRT, finite groups, the stated Pell theorems, and the identified logical calibration. They can be carried out over ZFC's own ω and coded finite sequences, rather than only for externally standard numerals. After the factorial fixed-point proof and exact polynomial projections are composed with these contracts, ZFC proves existence of a polynomial zero equivalent to a coded ZFC refutation; negating gives the asserted consistency biconditional. Exact file checks verify the supplied finite circuits and coefficients. They do not prove consistency or replace any imported semantic theorem.

4.5 Order of soundness and completeness

In soundness, first establish the natural domain and read the literal arithmetic gates. The unconditional base-modulus rows exclude zero power bases, and all exponents are syntactically positive. Apply the shared power theorem, then the common-radix theorem. Only then use the fixed-point theorem to identify F_H and the shared-radius estimate to identify F_t. The bounded CRT, beta, carrier, and flow arguments now recover C222 and the calibrated refutation. In particular, the request F_H^t does not assume $F_H = H!$ in order to prove it.

In completeness, start from a finite proof. Choose its outer row and label bounds, carrier, accumulator path, and beta data; these determine H. Set the actual factorials $F_H = H!$ and $F_t = t!$ and their shared radius. Next choose the CRT column and index representatives, then the common binomial radix, then the true values of the finitely many powers. Finally choose the common Pell frame and index auxiliaries. The polynomial equations contain mutual references, but this construction of witnesses has no unresolved cycle.

5 From natural zeros to integer zeros with fewer guards

The square-radius parent has 781 natural coordinates and 677 quadratic residuals. Guarding each coordinate independently with four squares would be wasteful: many signs are forced by rows

already present. The construction instead supplies a finite, ordered sign derivation for every original coordinate. There is no inference from a variable's name or intended interpretation.

5.1 An exact ternary guard

Define, for unrestricted integers a, b, c,

$$\mathcal{H}(a, b, c) = a^2 + b^2 + c(c + 1).$$

This is nonnegative because $c(c + 1) \geq 0$ for every integer c. Conversely, every $n \in \mathbb{N}$ is represented. Legendre's three-square theorem applies to $4n + 1$, which is not of the excluded form $4^r(8s + 7)$. Modulo four, exactly one of the three square roots is odd. Write the roots as $2a, 2b, 2c + 1$ and divide by four to obtain $n = a^2 + b^2 + c(c + 1)$. This representation is also explicitly formalized in the AFP entry *Three Squares* [5].

Appending $x = \mathcal{H}(a, b, c)$ is therefore an exact existential nonnegativity guard. It adds three integer unknowns and one quadratic row; it does not substitute a sum of squares into every occurrence of x and increase their degrees.

5.2 Four sign rules

Nonnegative polynomial definitions. Suppose a literal row says $x = f$, where f has nonnegative integer coefficients. Each monomial is nonnegative if every variable occurring to an odd exponent in it has already been shown nonnegative. Even powers need no sign assumption. The checker compares the complete coefficient dictionary of the defining row and checks precisely these dependencies.

Lemma 5.1 (One guard for two coordinates)**.** *For $x, y \in \mathbb{Z}$ with $x + y \geq -1$,*

$$x \geq 0 \text{ and } y \geq 0 \quad \longleftrightarrow \quad 2xy + x + y \geq 0.$$

Proof. The forward implication is immediate. Conversely, $(2x + 1)(2y + 1) = 2(2xy + x + y) + 1 > 0$. The two odd factors have the same sign. If both were negative, then $x, y \leq -1$, contradicting $x + y \geq -1$. Thus both are positive and $x, y \geq 0$. □

When a literal sum row proves $x + y \geq -1$, append the single row $2xy + x + y = \mathcal{H}(a, b, c)$. This is still quadratic and proves both signs. It is not an attempt to make two independent square representations share witnesses; it represents one nonnegative centered product.

Lemma 5.2 (A shifted quotient)**.** *If $A \geq 0$, $B \geq 1$, $q \in \mathbb{Z}$, and $A(q + 1) = B$, then $q \geq 0$.*

Proof. If $q \leq -1$, the left side is nonpositive, a contradiction. □

For a binomial digit certificate, the literal rows unfold to

$$N = d(Qh + o) + r, \qquad d = r + s + 1, \qquad Q = o + t + 1.$$

They give

$$dQ(h + 1) = N + dt + s + 1. \tag{10}$$

With all displayed dependencies nonnegative, Lemma 5.2 proves $h \geq 0$ without a guard. For residuals $R_1 = N - d(Qh + o) - r$, $R_2 = d - r - s - 1$, and $R_3 = Q - o - t - 1$, the difference of the two sides in (10) is exactly $-R_1 + dR_3 + R_2$. Euclidean division has the analogous identity $d(q + 1) = N + s + 1$. The checker expands and compares these finite identities, including the necessary arithmetic definitions.

Lemma 5.3 (Orient a root before Pell classification)**.** *Suppose the actual common frame gives*

$$W = 2 + \sum \textit{nonnegative atoms}, \qquad A^2 = 1 + W^3(W+2)(z+1)^2,$$

with $A, z \geq 0$*. Then* $A > W^2$*. If* $n \leq W$ *and*

$$X + Yn = YA + m + Tq$$

with $Y, m, T, q \geq 0$*, then* $X \geq 0$*.*

Proof. Here $W \geq 2$ and $(z+1)^2 \geq 1$, so $A^2 \geq 1 + W^3(W+2) > W^4$. As $A \geq 0$, $A > W^2 \geq n$. Therefore $X = Y(A-n) + m + Tq \geq 0$. □

This argument establishes the sign needed to apply the later Pell theorem; it does not assume the Pell classification while trying to prove one of its hypotheses. The certificate checks the exact frame norm, the sum defining W, the relevant base-modulus row, and a nonnegative-coefficient identity for $W - n$. All required signs occur earlier in its schedule.

5.3 The actual sign certificate

The frozen construction has 163 ordinary guards and eight paired guards. The latter seed 16 coordinate signs with only eight ternary representations. It then uses the rules in Table 3. All 781 original coordinates are proved nonnegative. Guard auxiliary coordinates remain unrestricted, as they should.

Rule applications after ordinary guard seeds	Number
Nonnegative polynomial evaluation	571
Shifted quotient	13
Centered pair (two conclusions each)	8
Common-frame root orientation	18

Table 3: Exact sign-certificate census. Counting conclusions gives $163 + 571 + 13 + 2 \cdot 8 + 18 = 781$.

There are $171 = 163 + 8$ ternary guard rows, giving $781 + 3 \cdot 171 = 1294$ unrestricted variables and $677 + 171 = 848$ residuals before projection.

Starting from an arbitrary integer zero, the guards and the ordered sign proof show that its original 781 coordinates form a natural zero of the unchanged parent. Conversely, a natural parent zero makes every selected coordinate and centered product nonnegative. The ternary representation extends it to an integer zero of the guarded system. This proves both existential directions without searching for solutions or assuming that every sign is syntactically obvious.

5.4 Certified monic projection

Over $\mathbb{Z}$, a row $v - f(y) = 0$ can be removed by substituting f for v in every other row. The inverse uniquely restores $v = f(y)$, regardless of the signs of the coefficients of f. Each step checks that the chosen row is exactly monic, that f does not contain v, and that *every* remaining residual stays quadratic. Guards are substituted along with all other rows and are never discarded merely because their variable has been eliminated.

For P_{956}, 338 coordinates and rows disappear: 223 affine restorations and 115 quadratic restorations, comprising 271 arithmetic gates and 67 other witnesses. Thus

$$1294 - 338 = 956, \qquad 848 - 338 = 510.$$

For P_{989}, restricting the support of a candidate restoration to at most four terms leaves 305 valid eliminations and therefore 989 variables and 543 rows. The support threshold controls the size of the resulting expression, not the mathematical validity of the projection.

The final sum of squares is expanded with exact integer arithmetic. The collected dictionary contains degree-four terms and every counted coordinate occurs. This proves the advertised exact degree, support, and height. Composing the preceding two-way transformations with the direct source calibration proves Theorem 1.1.

6 A short local majorant for nine positive variables

A separate reduction gives two explicit nine-positive-variable equations:

$$(442,\ 1\,520\,186), \qquad (456,\ 1\,376\,378),$$

where the entries are live binary arithmetic operations and exact total degree. The second improves both measures over the reviewed 475-operation, degree-1 379 730 equation with the same original31 tuple. The first is a tradeoff against the 433-operation, degree-1 628 768 compact54-sharp equation. Neither changes the nine-variable bound. The new construction is a short local Laurent majorant combined with a permissible successor coordinate. No global minimality or historical priority is asserted.

6.1 The local lemma

Let $A \geq 2$, $C \geq 2$, $i \geq 1$ be integers and put

$$\begin{aligned} &J = 4A - 5, && \delta = A^2 - 1, && D = 1 + \delta C^2, \\ &e = iJC^2, && F = 1 + \delta De^2, && T = iC^2\big[AJ(2D - C^2) + 3C^2 - 4\big]. \end{aligned}$$

Then

$$T > 2\sqrt{DF} > 0. \tag{11}$$

To prove this, set $t = C^2 \geq 4$. Exact expansion gives

$$T^2 - 4DF = i^2t^2Q(A,t) - 4D,$$

where

$$\begin{aligned} Q(A,t) = {} & 20t(3t-4)A^3 - t(83t-116)A^2 \\ & - 10(7t^2 - 14t + 8)A + 109t^2 - 224t + 116. \end{aligned}$$

Writing $A = x + 2$, $t = y + 4$, the coefficients of $Q(x+2, y+4) - (x+2)^2$ are

	y^0	y^1	y^2
x^0	1344	816	117
x^1	3580	2188	318
x^2	2975	1852	277
x^3	640	400	60

and are all positive. Consequently $Q(A,t) > A^2$ and

$$T^2 - 4DF > t^2A^2 - 4\big[(A^2-1)t+1\big] = A^2t(t-4) + 4t - 4 \geq 12.$$

The defining expression gives $T > 0$, since $2D - t = 2 + (2A^2-3)t > 0$, $J > 0$, and $3t - 4 > 0$. This proves (11).

The boundary condition is essential: at $(A,C,i) = (3,1,1)$ one obtains $T^2 - 4DF = -308$. It is therefore supplied before the final equation is assumed, by replacing its carrier input with $C = c+1$ for a positive unknown c. Canonical Pell witnesses already have $C > 1$, so this restriction preserves existential completeness.

6.2 The complete polynomial

The nine positive unknowns are $b, g, h, s, w, c, i, j, n$. The two inherited source prefixes define R, N, cap from their complete literal coefficient tables. The delivered package includes both tables, all four-port numerals, all fixed circuit leaves, and the exact prefix maps. The whole compact54-sharp and original31 tuples are kept separate. Both source prefixes prove $R, N \geq 8$ and $b \leq R, N$ on the ambient positive domain.

Set $C = c+1$ and retain

$$\begin{aligned}
&U = N^2w, && Y = N^2s, && M = RY, && P = 2M^2U,\\
&K = R + 1 + h(P-1), && A = M(U+1), && B_1 = 2R+1, && W_1 = bw,\\
&J = 4A - 5, && Q = 3W_1C - 2(W_1^2 - 1),\\
&a_0 = 1 + (P^2-1)K^2, && V_0 = PK,\\
&m = K^2(4\,\mathrm{cap} - g) - 16\,\mathrm{cap}(C - KY)^2,\\
&\Omega = Jm, && E_0 = J(n + W_1^2) + Q.
\end{aligned}$$

For D, F, e as above define

$$\begin{aligned}
&\Lambda = CD + 1 - A, && G = A + F\Lambda,\\
&Z = B_1 + (j-1)C, && H = C + FZ, && I = 1 + (G^2-1)H^2,\\
&V_1 = TGH, && a_1 = 4DFI, && \Theta = E_0 - \Omega(V_0 + V_1).
\end{aligned}$$

The equation is the ordinary integer polynomial

$$\big[\Theta^2 + \Omega^2(a_0 - a_1)\big]^2 - 4\Omega^2 a_0 \Theta^2 = 0. \tag{12}$$

All named quantities are polynomial abbreviations, and add no unknowns. The circuit counts charge the successor $C = c+1$, scalar products, and all squares. There are no divisions or variable exponents.

6.3 The unbounded equivalence proof

The source bounds give $A \geq 33280$. Also $D, F, G, H > 0$, $G > A$, and $H > C \geq 2$. Therefore $I < (GH)^2$, and the new lemma gives $V_1 > \sqrt{a_1}$. Independently, $V_0^2 - a_0 = K^2 - 1 > 0$, so $V_0 > \sqrt{a_0}$. These inequalities hold before a zero is assumed.

At a zero, $E_0 = Jn + (J-2)W_1^2 + 3W_1C + 2 > 0$ excludes $\Omega = 0$. Factoring the real two-root norm yields signed roots $r_0^2 = a_0$, $r_1^2 = a_1$ with

$$E_0 = \Omega(V_0 + V_1 + r_0 + r_1).$$

Both strict majorants make the bracket positive for all sign choices. Hence $m > 0$. Only now do the inherited margin estimates give $|C - KY| < K/2$, $C > K$, and $C > B_1 + W_1$. Moreover $A^2 > P$, $D > C^2$, $F \geq A^2$, and $I > A^2$, whence

$$DFI > C^2A^4 > K^2P^2 > a_0.$$

Thus $x = r_0 + r_1 = \Theta/\Omega$ is nonzero and rational. The isolation identity $2xr_0 = x^2 + a_0 - a_1$ makes both roots rational, hence integral. The rational number $r_1/2$ has integer square DFI, so it too is integral. The norm relation now gives $J \mid Q$.

The retained saturated Pell theorem recovers $C = \psi_A(B_1)$ from the square DFI; all its range hypotheses have now been established. Together with a_0 square, $m > 0$, and $J \mid Q$, these are the exact inherited ratio conditions, which recover the encoded source zero.

Conversely, a source zero supplies canonical ratio witnesses and compatible saturated-Pell auxiliaries with $W_1^2 + Q/J < 2D$. Its carrier satisfies $C = \psi_A(B_1) > 1$, so $c = C - 1$ is positive. Choose positive roots $u_0 = \sqrt{a_0}$ and $u_1 = 2\sqrt{DFI}$, and define

$$n = m(V_0 + V_1 + u_0 + u_1) - W_1^2 - Q/J.$$

This is an integer. Since $F > D$, the lemma gives $V_1 > 2D$, so $n > 0$. Then $\Theta = \Omega(u_0 + u_1)$ and (12) holds. This proves both existential directions. Negation and the parents' retained source/refutation calibration give the Con(ZFC) biconditional, with neither consistency nor inconsistency assumed. No new Lean, Rocq, or object-ZFC proof term is claimed.

6.4 Cancellation-resolved exact degree

Let $r = \deg R$, $\nu = \deg N$, $a = r + 4\nu + 2$, and $W = 116$. The unchanged source has an isolated positive leading kernel coefficient. The shared degrees are

$$\deg \Omega = 5a + 2W + 3, \qquad \deg V_0 = 3a + r + 1,$$
$$\deg E_0 = a + 4, \qquad \deg \delta_0 = 3a + r + 2,$$

where $\delta_0 = V_0^2 - a_0$. In weights $(a, 1, 1, r)$ on (A, C, i, Z), the unique leading monomials are

$$\begin{aligned}
[T]_{\text{top}} &= 8A^4C^4i, \\
[T^2 - 4DF]_{\text{top}} &= 60A^3C^8i^2, \\
[GH]_{\text{top}} &= 256A^{14}C^{15}i^4Z, \\
[V_1]_{\text{top}} &= 2048A^{18}C^{19}i^5Z, \\
[\delta_1]_{\text{top}} &= 3932160A^{31}C^{38}i^{10}Z^2,
\end{aligned}$$

where $\delta_1 = V_1^2 - a_1$. The last line follows from

$$\delta_1 = (T^2 - 4DF)(GH)^2 + 4DF(H^2 - 1).$$

Thus $\deg V_1 = 18a + r + 24$ and $\deg \delta_1 = 31a + 2r + 48$. Set

$$B_* = E_0^2 - 2E_0\Omega(V_0 + V_1) + \Omega^2(\delta_0 + \delta_1).$$

The exact outer identity is

$$F_9 = B_*^2 + 4\Omega^2 V_0 V_1 B_* + 4\Omega^4(\delta_0 V_1^2 + a_0\delta_1).$$

Its four summand degrees are

$$82a + 4r + 8W + 108, \quad 72a + 4r + 8W + 85, \quad 59a + 3r + 8W + 62, \quad 57a + 4r + 8W + 62.$$

The first is strictly greatest and has positive nonzero leading coefficient. The exact degree is therefore

$$86r + 328\nu + 8W + 272.$$

For $(r, \nu) = (8799, 2324)$ this is 1 520 186; for $(r, \nu) = (7943, 2110)$ it is 1 376 378. The independent checker also computes nonzero specialized leading coefficients modulo three primes using the actual fixed numerals. Those coefficient witnesses supplement the universal identity and unique-leader argument.

6.5 Validation and comparison

The independent standard-library verifier imports no builder. It verifies all fixed numerals and their recipes, complete prefix maps, every new register, local positivity and norm identities, topology, liveness, exact operation counts, degree attainment, and full deterministic regeneration. Normal and optimized executions agree. It rejects the invalid $C = 1$ boundary and an altered majorant gate.

The operation breakdown is

	+	−	×	total
compact2082	159	22	261	442
H1868	155	22	279	456

with exactly nine positive variables in each case. The whole coefficient bindings and inherited ordinary proofs remain explicit. Finite circuit checks are not substituted for the unbounded native/compiler/refutation calibration, and no global simplicity optimum is claimed.

7 Shorter exact radical norms

The final contribution changes only an evaluator, not its polynomial or zero set. It consequently transfers the parent's exact degree and semantic equivalence without a new witness transformation. All actual fixed numeral payloads are retained.

Define the formal radical norm

$$F_m(x; a_1, \ldots, a_m) = \prod_{\varepsilon \in \{\pm 1\}^m} \left(x + \sum_{j=1}^{m} \varepsilon_j \sqrt{a_j} \right).$$

Conjugation shows that this is an integer polynomial in $x, a_1, \ldots, a_m$. To handle one rational radicand, use the cleared polynomial

$$\delta^{2^{m-1}} F_m(x; a_1, \ldots, a_{m-1}, C/\delta).$$

The following are identities of integer polynomial rings, including at $\delta = 0$ after clearing. Their literal circuits have neither radicals nor division. Any nonzero-denominator hypothesis in the parent's semantic interpretation is unchanged.

7.1 A polarized three-root norm

Set

$$z = x^2, \qquad G = \delta(z + a - b) - C, \qquad U = \delta^2 az, \qquad V = \delta bC.$$

Then

$$\delta^4 F_3(x; a, b, C/\delta) = [G^2 + 4(U - V)]^2 - 16UG^2. \tag{13}$$

One can verify this by taking the two conjugates of the two-root norm, or by comparison with the symmetric form

$$[J^2 - 4(T + R(a + b))]^2 - 64TRx^2,$$

where $J = \delta(x^2 - a - b) - C$, $T = \delta^2 ab$, and $R = \delta C$. The exact checker compares all 35 coefficients in the free variables x, a, b, C, δ. Sharing the displayed products gives 18 operations, one fewer than the retained 19-operation implementation.

7.2 A four-root norm with shared inner products

Let $s = a + b$, $x_2 = x^2$, and define

$$\begin{aligned} q_0 &= \delta(x_2 + c - s) - C, & q_1 &= -2\delta x, & & \\ T &= \delta^2 ab, & R &= \delta C, & z &= 64TR, \\ A_0 &= q_0^2 + cq_1^2 - 4(T + Rs), & A_1 &= 2q_0 q_1, & & \\ f_0 &= A_0^2 + cA_1^2 - z(x_2 + c), & f_1 &= 2(A_0 A_1 + zx). & & \end{aligned}$$

In the quadratic algebra $u^2 = c$, the inner cleared three-root norm at $x - u$ is $f_0 + f_1 u$. Its norm is therefore

$$f_0^2 - cf_1^2.$$

This uses 38 operations rather than 43. All 495 free polynomial coefficients are compared. The archive supplies the generic identity; it does not claim an additional complete four-root endpoint in the comparison table.

7.3 Pair diagonal conjugates in the five-root norm

Work in the biquadratic algebra $u^2 = c$, $v^2 = d$. If the inner cleared norm is written

$$f = f_0 + 2f_1 u + 2f_2 v + 2f_3 uv,$$

pair simultaneous sign reversal first. Direct multiplication gives

$$f(u, v) f(-u, -v) = G + 4uvH,$$

where

$$G = f_0^2 + 4(cdf_3^2 - cf_1^2 - df_2^2), \qquad H = f_0 f_3 - 2f_1 f_2.$$

The other two conjugates give $G - 4uvH$. Thus

$$\mathrm{Norm}(f) = G^2 - 16cdH^2. \tag{14}$$

Keeping the odd components divided by two is the useful computational choice: there is no need to materialize all three factors of two before taking the norm.

For completeness, the inner components are supplied by the following finite polynomial formulas. Set $t = x + u + v$, $s = c + d$, and

$$\begin{aligned}
q_0 &= \delta(x^2 + s - a - b) - C, & q_3 &= 2\delta, & q_1 &= q_3 x,\\
T &= \delta^2 ab, & R &= \delta C, & z &= 64TR,\\
A_0 &= q_0^2 + sq_1^2 + cdq_3^2 - 4(T + R(a+b)),\\
A_1 &= 2q_1(q_0 + dq_3), & A_2 &= 2q_1(q_0 + cq_3),\\
A_3 &= 2(q_0 q_3 + q_1^2).
\end{aligned}$$

Then

$$\begin{aligned}
f_0 &= A_0^2 + cA_1^2 + dA_2^2 + cdA_3^2 - z(x^2 + s),\\
f_1 &= A_0 A_1 + dA_2 A_3 - zx,\\
f_2 &= A_0 A_2 + cA_1 A_3 - zx,\\
f_3 &= A_0 A_3 + A_1 A_2 - z.
\end{aligned}$$

Indeed $q = \delta(t^2 - a - b) - C = q_0 + q_1 u + q_1 v + q_3 uv$, and these are exactly the components of the cleared three-root polynomial $[q^2 - 4(T + R(a+b))]^2 - zt^2$.

The resulting five-root circuit uses 84 operations rather than 93. The checker compares all 20,349 coefficients of the free norm identity and verifies exact substitution into the seven complete boundary registers of each 13-integer parent. This proves equality of the *whole* parent and child polynomials, not agreement on sampled inputs. It gives the four 13-variable improvements in Table 2; (13) gives the 11-positive-variable improvement.

8 Literal specification, provenance, and proof status

8.1 What makes these equations explicit

The direct and integer equations are supplied twice: as literal quadratic residual arrays and as fully collected quartic coefficient dictionaries. A term is encoded as

```
[coefficient, [zero_based_variable_indices]]
```

Repeated indices denote powers; an empty index list denotes a constant. For example, $-3X_2^2X_7$ is `[-3,[2,2,7]]`. Every coefficient is a literal signed integer. Variable names are annotations and do not alter the polynomial. The archive's manifest authenticates every byte.

The small-variable equations are literal finite straight-line programs with addition, subtraction, and multiplication. Every fixed leaf is bound to the included exact numeral bytes, separately for each coherent source tuple. They are not parameterized families awaiting a choice of coefficients. Radical, factorial, power, finite-product, and proof-calculus notation occurs only in explanations and proofs.

Printing 10,988 monomials in the article would obscure the definition without making it more exact. The coefficient file, its residual definition, the deterministic generator, and the independent expansion checker give four mutually checkable specifications. The full file is part of this submission, not an external dependency or an unspecified computable constant.

8.2 Important entry points in the archive

Path	Content
`README.md`	Results, exact domains, commands, and proof scope.
`report/main.tex` and `report/main.pdf`	This article, with its TeX inputs.
`direct/equations/parent781/`	Frozen square-radius natural parent, macro ledger, and bound certificates.
`direct/equations/natural467/`	Complete output-linear natural quartic and restoring maps.
`direct/equations/natural468/`	Complete square-radius natural quartic.
`direct/equations/natural469/`	Smaller-support natural alternative.
`integer/equations/integer956/`	Main integer quartic, guarded parent, sign proof, restoring maps, and expansion.
`integer/equations/integer989/`	Integer alternative with 26-bit coefficient height.
`nine/equations/`	Two complete nine-positive-variable polynomial DAGs.
`norms/equations/`	Five complete shorter polynomial evaluators.
`direct/provenance/upstream/`	Identified inherited direct arithmetic and source-calibration materials.
`BASELINES.md`	Exact comparison sources at the pinned repository snapshot.
`verify.py`	Portable release checks, including all component checks.
`checks/`	Release receipts and independent review scopes.
`MANIFEST.json`	SHA-256 inventory of the delivered files.

8.3 What the exact checks establish

The release checks rebuild both natural parents byte for byte, reproduce the natural contractions, replay every restoring substitution, and independently collect every squared residual. They authenticate source pins, actual fixed numeral bytes, and constructor and macro bindings. The integer checker imports neither its builder nor its projection optimizer: it independently verifies sign dependencies, coefficient identities, guard coverage, projections, and the final expansion. It rejects a missing pair guard, a false frame margin, an altered restoring polynomial, and an altered final coefficient.

The nine-positive-variable checker verifies the exact majorant defect, its positivity certificate, all new register formulas, norm identities, literal source-prefix bindings, liveness, operation counts, and the cancellation-resolved exact degree. Its nonzero leading coefficients are also authenticated for the actual numeral specialization. The norm checker proves free polynomial identities by coefficient comparison and authenticates their substitutions into each complete parent.

Normal Python and optimized Python runs are supplied. The mathematical checks use explicit exceptions, not removable `assert` statements as their acceptance condition. The component executables need only the Python standard library and local packaged files. No verification command searches for a solution of a consistency equation.

8.4 What the checks do not establish

Finite algebra checks do not replace the unbounded equivalence proofs. The factorial fixed-point proof, source reparameterization, integer sign rules, and nine-variable majorant argument are ordinary mathematical proofs formalizable in ZFC. Their interpretation as consistency equations also uses the identified inherited compiler, finite-flow, and C222-to-ZF-to-ZFC calibration theorems.

This release does not claim a new complete Lean or Rocq proof term, nor a fully emitted derivation in an encoded object-language ZFC proof system. The retained Lean sources identify the calibration interfaces; the release's standard-library checkers do not invoke a proof-assistant kernel or rebuild the entire upstream formal development. A reader wanting that stronger assurance must formalize the new transformations and replay the identified upstream development.

The distinctions are material: a correct coefficient transcript, a universal mathematical equivalence, a proof-assistant theorem, and an encoded internal proof object are different deliverables. This archive supplies the first two and identifies the inherited dependencies of the second. It does not label an experimental receipt as a proof of Con(ZFC).

8.5 Reproduction

From the extracted archive root, run

```
python3 verify.py
python3 -O verify.py
sh build_report.sh
```

The first two commands run the packaged checks in a temporary copy, so the verified delivered files are not rewritten. The report build requires a conventional LaTeX installation with the packages listed in `report/main.tex`. Mathematical validation does not require LaTeX or internet access.

For a particular integer equation, the independent entry point is

```
python3 integer/verify_integer.py \
  --parent direct/equations/parent781/equation.json \
  --equations integer/equations/integer956
```

For reconstruction, `integer/build_integer.py` accepts the literal natural parent and an output directory; the support caps 16 and 4 give the two delivered integer variants. The source compiler accepts `-radius square` or `-radius linear`.

9 Assessment of the contribution

The most reusable result is Theorem 2.2. A large independent precision parameter is often introduced in Diophantine approximations to an integer-valued function. Here a polynomial in the proposed output supplies that precision, and a two-stage lower bound proves uniqueness of the fixed point. The gain is not just the replacement of one displayed formula by another: one actual positive-power request, its distinct base-modulus machinery, and associated coordinates disappear from the emitted polynomial.

The rule-branch reduction attacks a different source of size: two labels were kept as local bounded coordinates even though the selected inference rule already determined them. Eliminating them before the bounded-quantifier compiler saves more than eliminating two variables at the end. The enlarged carrier label bound is essential to that argument and is included explicitly.

The integer reduction then uses consequences of the literal rows to avoid independent sign representations. The shifted-quotient and common-frame arguments are semantic facts about an integer zero of those rows; exact identities and an acyclic dependency schedule make those facts independently checkable. The resulting equation remains quartic throughout.

The smaller-variable improvements pursue a separate goal. The nine-variable construction obtains a stricter majorant with controlled cost and exact degree, while the norm identities preserve entire polynomials and reduce their evaluation cost. These are concrete improvements over specified parents, with all coefficient choices fixed. None establishes a globally simplest consistency equation or certifies that the ideas have never appeared elsewhere.

References

[1] Vladimir Reshetnikov and contributors, *ProveIt*, repository snapshot `8f8fac484b976ba0ab75fd173282e838ca5a3b85`. https://github.com/VladimirReshetnikov/ProveIt/tree/8f8fac484b976ba0ab75fd173282e838ca5a3b85. Relevant directories: `Computability/HilbertTenthProblem`, `Computability/DiophantineConsistency/con-zfc-diophantine-forms`, and `docs/incoming`.

[2] ProveIt incoming submission, *ConZFC Synchronized Certificates*, `ConZFC_Synchronized_Certificates_2026-10-24.zip`, at the snapshot in [1]. The present direct parent imports its shared positive-power frame, product-index sharing, and direct source lineage.

[3] ProveIt incoming submission, *ConZFC batched certificates*, `con_zfc_batched_certificates_2026-10-23.zip`, at the snapshot in [1]. Source of the 1,690-integer direct baseline, the degree-1,072 13-integer parent, and the monic projection adapter.

[4] ProveIt incoming submission, *ConZFC 13 Integer Norms*, `ConZFC_13_Integer_Norms_2026-10-24.zip`, at the snapshot in [1]. Exact retained parents and notices are included under `norms/upstream/`.

[5] Anna Danilkin and Loïc Chevalier, *Three Squares*, Archive of Formal Proofs, 3 May 2023. https://isa-afp.org/entries/Three_Squares.html. Includes the integer nonnegativity representation $n = x^2 + y^2 + z^2 + z$.

[6] James P. Jones, *Universal Diophantine equation*, Journal of Symbolic Logic **47** (1982), 549–571. Background for the separate universal-polynomial line; it is not an ingredient of the new direct C222 quartic source.

[7] Jonas Bayer, Marco David, Malte Haßler, Yuri Matiyasevich, and Dierk Schleicher, *Diophantine equations over $\mathbb{Z}$: universal bounds and parallel formalization*, arXiv:2506.20909, version 3, 22 October 2026. https://arxiv.org/abs/2506.20909. Background on keeping variable, degree, coefficient, and formalization claims separate.